Project 2.2.7 Creating a Comand GUI

Project 2.2.7 — My Plan

What the program needs to do: The program opens a window where you type in a website or IP address, click a button to run a network command, and see the results on screen. You can also save those results to a file.

Step 1 — Set up the window

Create a basic tkinter window with a title. Add a frame inside it to hold everything neatly.

Step 2 — Add a text box for the domain

Put a label that says "Enter Domain or IP" and an entry box below it. Pre-fill it with www.pltw.org so the user knows what format to type in.

Step 3 — Create the three command buttons

Add three buttons — Ping, Traceroute, and NSLookup. Each button calls the same do_command() function but passes a different command name using lambda so the function knows which one to run.

Step 4 — Write the do_command function

This function grabs whatever the user typed in the entry box. If it's empty, use localhost. Then it builds the right command for the operating system (Windows uses tracert, Mac/Linux uses traceroute). It runs the command, and prints the output line by line into the text box so you can watch it in real time.

Step 5 — Add the output text box

Put a scrollable text box at the bottom of the window. This is where all the command results show up as they run.

Step 6 — Add a Save button

Write a mSave() function that opens a "Save As" dialog, lets the user pick a filename, then writes everything from the output text box into that file.

Step 7 — Test it

Run the program, type in a domain, and click each button one at a time. Make sure output shows up. Then hit Save and check that the file was created correctly.

START

IMPORT subprocess, tkinter, and filedialog tools

CREATE window titled "Network Tools" with a frame inside it

ADD label "Enter Domain or IP" and entry box pre-filled with "www.pltw.org"

ADD Ping button that calls do_command("ping") when clicked

ADD Traceroute button that calls do_command("tracert") when clicked

ADD NSLookup button that calls do_command("nslookup") when clicked

ADD Save button that calls mSave() when clicked

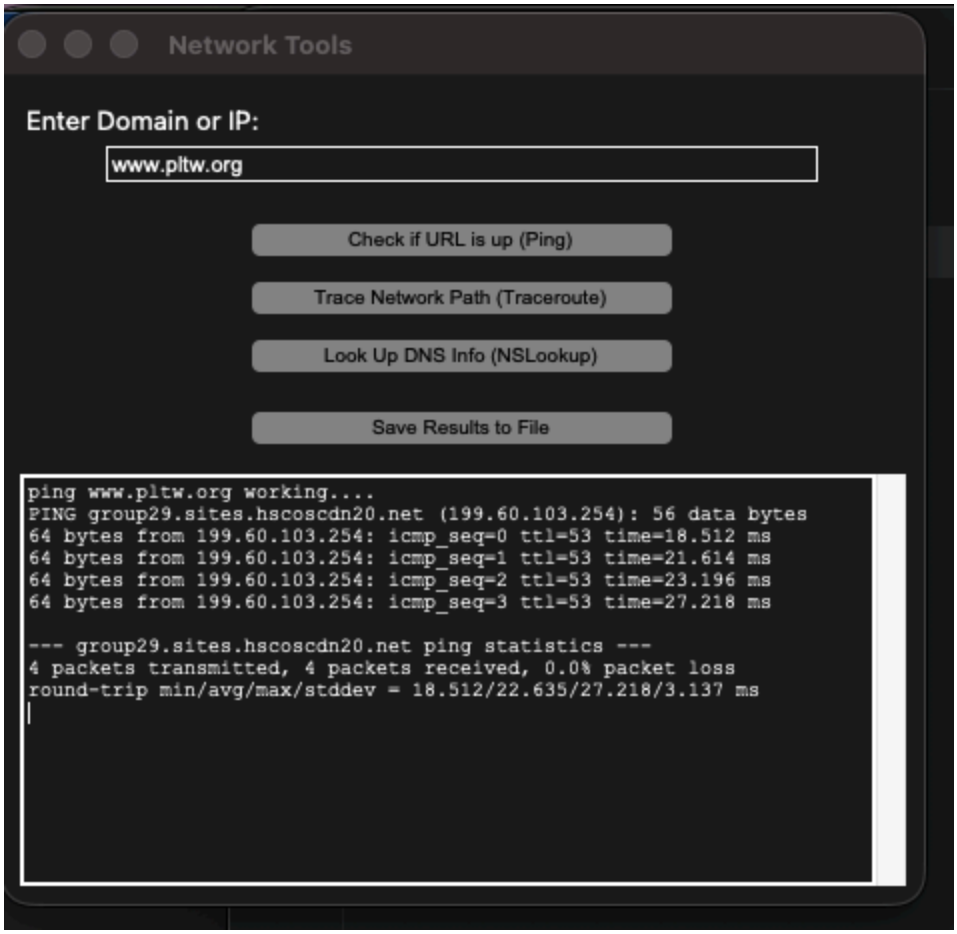
ADD scrollable output text box at the bottom of the window

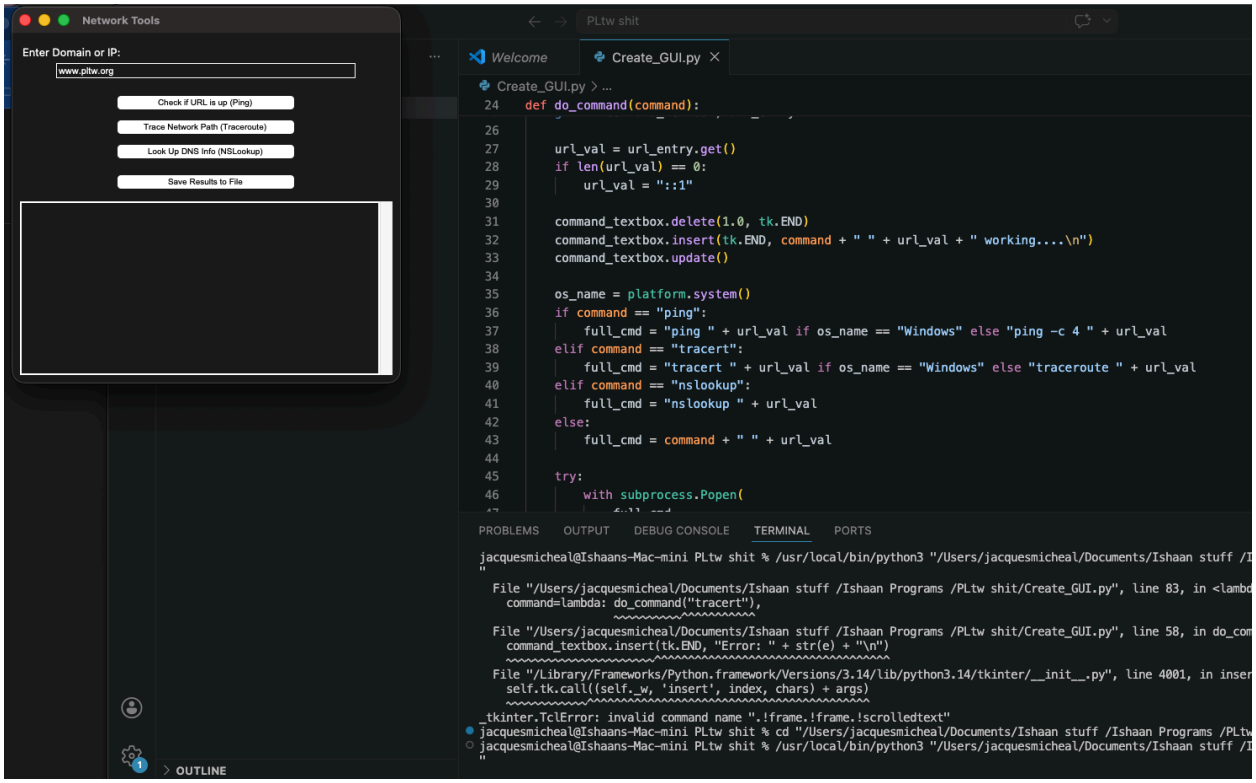
WHEN do_command(command) is called GET text from the entry box IF entry box is empty SET url to localhost CLEAR the output box DISPLAY command + "working...." CHECK operating system IF Windows BUILD "tracert" ELSE BUILD "traceroute" IF ping BUILD "ping" IF nslookup BUILD "nslookup" RUN the command WHILE output is coming in PRINT each line to the output box UPDATE window in real time IF error occurs DISPLAY error message in output box

WHEN mSave() is called OPEN Save As dialog IF user picked a filename OPEN file for writing GET all text from output box WRITE text to file CLOSE file IF user cancelled DO nothing and return

WAIT for user input

END


Network Tools
Enter Domain or IP:
www.pltw.org
Check if URL is up (Ping)
Trace Network Path (Traceroute)
Look Up DNS Info (NSLookup)
Save Results to File
ping www.pltw.org working....
PING group29.sites.hscoscdn20.net (199.60.103.254): 56 data bytes
64 bytes from 199.60.103.254: icmp_seq=0 ttl=53 time=18.512 ms
64 bytes from 199.60.103.254: icmp_seq=1 ttl=53 time=21.614 ms
64 bytes from 199.60.103.254: icmp_seq=2 ttl=53 time=23.196 ms
64 bytes from 199.60.103.254: icmp_seq=3 ttl=53 time=27.218 ms
--- group29.sites.hscoscdn20.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 18.512/22.635/27.218/3.137 ms


Network Tools
Enter Domain or IP:
www.pltw.org
Check if URL is up (Ping)
Trace Network Path (Traceroute)
Look Up DNS Info (NSLookup)
Save Results to File
PLtw shit
Welcome
Create_GUI.py
Create_GUI.py > ...
24 def do_command(command):
27 url_val = url_entry.get()
28 if len(url_val) == 0:
29 url_val = "::1"
31 command_textbox.delete(1.0, tk.END)
32 command_textbox.insert(tk.END, command + " " + url_val + " working....\n")
33 command_textbox.update()
35 os_name = platform.system()
36 if command == "ping":
37 full_cmd = "ping " + url_val if os_name == "Windows" else "ping -c 4 " + url_val
38 elif command == "tracert":
39 full_cmd = "tracert " + url_val if os_name == "Windows" else "traceroute " + url_val
40 elif command == "nslookup":
41 full_cmd = "nslookup " + url_val
42 else:
43 full_cmd = command + " " + url_val
45 try:
46 with subprocess.Popen(
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
jacquesmicheal@Ishaans-Mac-mini PLtw shit % /usr/local/bin/python3 "/Users/jacquesmicheal/Documents/Ishaan stuff /I
File "/Users/jacquesmicheal/Documents/Ishaan stuff /Ishaan Programs /PLtw shit/Create_GUI.py", line 83, in <lambd
command=lambda: do_command("tracert"),
File "/Users/jacquesmicheal/Documents/Ishaan stuff /Ishaan Programs /PLtw shit/Create_GUI.py", line 58, in do_com
command_textbox.insert(tk.END, "Error: " + str(e) + "\n")
File "/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/tkinter/__init__.py", line 4001, in inser
self.tk.call((self._w, 'insert', index, chars) + args)
_tkinter.TclError: invalid command name ".!frame.!frame.!scrolledtext"
jacquesmicheal@Ishaans-Mac-mini PLtw shit % cd "/Users/jacquesmicheal/Documents/Ishaan stuff /Ishaan Programs /PLtw
jacquesmicheal@Ishaans-Mac-mini PLtw shit % /usr/local/bin/python3 "/Users/jacquesmicheal/Documents/Ishaan stuff /I
OUTLINE